Ioannis P. Rizos

publishing as *J. Rizos*

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Professor of Theoretical Physics at the University of Ioannina since 2014. Author of **71 papers in international peer-reviewed journals** (5,173 citations, h-index 36); Marie Curie Fellow (1996, 2003) and Scientific Associate at CERN (2013). Thirty years of teaching, examining and curriculum design in theoretical physics — quantum mechanics, classical physics, electrodynamics — and in computational methods, including four years on the Hellenic Open University's M.Sc. in Data Science and Machine Learning. Former Head of the Department of Physics (2013-2017) and President of the Southeastern European Network in Mathematical and Theoretical Physics (SEENET-MTP) since 2018.

Research Interests

String phenomenology and the construction of realistic four-dimensional string vacua; classification of heterotic superstring models in the free-fermionic and orbifold formulations; non-supersymmetric string model building and the decompactification problem; grand unified theories, neutrino masses and proton stability; string effective actions, higher-curvature gravity and non-singular cosmological solutions. Computational and machine-learning approaches — genetic algorithms, quantum annealing and classification techniques — applied to the exploration of the string landscape.

Contact

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Education

Nov 1991	**Ph.D. in Physics**, University of Ioannina. Thesis: *Construction and Phenomenological Analysis of Supersymmetric Unified Models from Four-Dimensional Superstring Theories.* Advisor: Prof. K. Tamvakis.
1986 – 1991	Doctoral researcher (Special Postgraduate Scholar, E.M.Y.), Department of Physics,

	University of Ioannina.
1984 – 1985	Graduate studies in Physics, University of Crete.
1980 – 1984	B.Sc. in Physics, Department of Physics, University of Ioannina.

Academic Appointments

2014 – present	**Professor**, Department of Physics, University of Ioannina (elected June 2014; appointed 4 September 2014).
2004 – 2014	**Associate Professor**, Department of Physics, University of Ioannina (elected November 2004; appointed June 2005).
1996 – 2004	**Assistant Professor**, Department of Physics, University of Ioannina — elected February 1994 while a postdoctoral researcher at SISSA/ISAS; took up duties October 1996 following military service and a further research appointment at SISSA; tenured April 2000.
1994 – 1995	Military service, Hellenic Army (Signal Corps).
1993 – 1994	Postdoctoral Researcher, International School for Advanced Studies (SISSA/ISAS), Trieste, Italy.
1991 – 1993	Postdoctoral Researcher, Centre de Physique Théorique, École Polytechnique, Palaiseau, France.

Visiting Positions

Jun – Sep 2013	**Scientific Associate**, CERN, Geneva, Switzerland.
Feb – Jul 2003	**Visiting Professor** (*Professeur Invité*), Centre de Physique Théorique, École Normale Supérieure, Paris, France (*Marie Curie Fellow*).
Oct 2000 – Jul 2001	**Visiting Professor** (*Professeur Invité*), Centre de Physique Théorique, École Polytechnique, Palaiseau, France.
May – Aug 2000	*Corresponding Associate*, CERN, Geneva, Switzerland.
Nov 1995 – Oct 1996	Visitor, SISSA/ISAS, Trieste, Italy (*Marie Curie Fellow*).
Oct – Nov 1995	Visitor, CERN, Geneva, Switzerland.

Fellowships and Awards

2003	Marie Curie Fellow, European Union (6 months).
1995 – 1996	Marie Curie Fellow, European Union (6 months).
1991 – 1993	Fellow of the European Community (*Science* Programme).
1986 – 1991	Special Postgraduate Scholar (E.M.Y.), Department of Physics, University of Ioannina.
1981 – 1984	Fellow of the Greek State Scholarships Foundation (I.K.Y.), three consecutive years of undergraduate study.

Research Funding and Projects

High Energy Physics

2019 – 2023	COST Action CA18108 *QG-MM: Quantum Gravity Phenomenology in the Multi-Messenger Approach*.
2012 – 2016	*ARISTEIA*: Particle Physics and Cosmology in the LHC Era, co-funded by the European Social Fund.
2012 – 2016	*THALES*: Theoretical Elementary Particle Physics and Cosmology in the Light of the LHC, co-funded by the ESF.
2012 – 2015	Particle Phenomenology and Cosmology in the LHC Era (K.E. 80906), NSRF 2007–13.
2009 – 2013	ITN *Unification in the LHC Era*, PITN-GA-2009-237920, European Union.
2006 – 2009	RTN *UniverseNet*, MRTN-CT-2006-035863, European Union.
2005 – 2007	*HERAKLEITOS*: Embedding of Unified Gauge Theories in Higher-Dimensional Spaces, EPEAEK.
2004 – 2009	RTN *The Quest for Unification: Theory Confronts Experiment*, MRTN-CT-2004-503369, European Union.
2001 – 2004	RTN *Supersymmetry and the Early Universe*, HPRN-CT-2000-00152, European Union.
2000 – 2004	RTN network *Across the Present Energy Frontier: Probing the Origin of Mass*, HPRN-CT-2000-00148, European Union — member of the research team and **scientific coordinator and representative of the Ioannina node**.
1997 – 1999	TMR *Beyond the Standard Model*, FMRX-CT96-0090, European Union.
1991 – 1993	*Science* Programme, European Union (fellow).

Information Technology, Networks and e-Learning — as Scientific Coordinator

Scientific Coordinator renders the Greek role Epistimonikos Ypefthynos*: the academic bearing scientific and administrative responsibility for a project before the university's Research Committee. These are ICT infrastructure and e-learning projects rather than research grants.*

2023 – 2026	**Scientific Coordinator**, *Extension and Development of the Greek School Network* (K.E. 83427), Ministry of Education.
2016 – present	**Scientific Coordinator**, *Acquisition of Academic Teaching Experience by Young Postdoctoral Scholars at the University of Ioannina*, OP Human Resources Development, Education and Lifelong Learning.
2012 – 2015	**Scientific Coordinator**, *Open Academic Courses of the University of Ioannina*, NSRF 2007–13.
2011 – 2015	**Scientific Coordinator**, *STIRIZO — Horizontal project supporting schools, teachers and pupils on the road to the digital school*, NSRF 2007–13.
2006 – 2009	**Scientific Coordinator**, *Advanced Telematic Services for Education and Research at the University of Ioannina*, OP Information Society.
2005 – 2008	**Scientific Coordinator**, *Epirus Web Portal* (K.E. 11967 design and implementation; K.E. 11968 equipment), Ministry of Education.
2004 – 2009	**Scientific Coordinator**, *Technical Support for the Construction of Metropolitan Broadband Fibre-Optic Networks in the Region of Epirus*, OP Information Society.
2004 – 2006	**Scientific Coordinator**, *Advanced Telematic Services for Secondary Education* (prefectures of Ioannina, Arta, Preveza, Thesprotia, Corfu and Lefkada), OP Information Society.

Information Technology and e-Learning — as participant

2001 – 2005	Advanced Telematic Services for Primary Education; School Information Systems Help Desk; *DIKTYOTHEITE* (EU, Information Society); equipment and upgrading of tele-teaching and computer laboratories — OP Information Society (various calls).
2001 – 2003	Advanced Telematic Services at the University of Ioannina; *EDUNET* Panhellenic Network for Education (Ioannina region); wireless connectivity for school units.
2003 – 2005	Equipment and Networking in Primary Education, OP Information Society.
2005 – 2008	New Postgraduate Programmes of the University of Ioannina — *New Technologies and Research in Physics Education*, Ministry of Education.
2007 – 2008	Training of Trainers Programmes (PAKE), Ministry of Education — teaching of General and Distance-Education methodology modules for the PE04 discipline (64 teaching hours).
2019	Teacher Training for the Exploitation and Application of Digital Technologies, Ministry of Education.
1999 – 2000	Distance Support for Mathematics Teaching through Networks and Computing Tools, Ministry of Education.

Teaching Experience

Hellenic Open University — M.Sc. in Data Science and Machine Learning (taught in English)

2021 – 2025	**DAMA50 — Mathematics for Machine Learning.** Tutor (S.E.P.), four academic years; **Module Coordinator** 2022–23 and 2023–24.

Hellenic Open University — B.Sc. Programme in Natural Sciences

2017 – 2021	**FYE40 — Quantum Physics.** Tutor, four academic years; Module Coordinator 2017–18 through 2021–22.
2004 – 2017	**FYE34 — Classical Physics II.** Tutor, thirteen academic years; Module Coordinator 2015–16 and 2016–17.
2013 – 2014	**FYE14 — Introduction to the Natural Sciences.** Module Coordinator.

University of Ioannina — Undergraduate courses, Department of Physics

2018 – present	Introduction to Quantum Field Theory (elective; 7 semesters).
2015 – present	Quantum Theory II (compulsory; 10 semesters).
2011 – 2017	Classical Electrodynamics I (compulsory; 5 semesters).
2004 – 2025	Classical Mechanics II (compulsory; 12 semesters).
2004 – 2011	Classical Mechanics I (compulsory; 7 semesters).
2002 – present	*Mathematics and Physics with Computers* (elective; designed and taught; 22 semesters).
2002 – 2011	Mathematics for Physicists (elective; 7 semesters).
2002 – 2011	*New Technologies in Education* (elective; designed and taught; 8 semesters).
2003 – 2005	*Internet Applications* (elective; designed and taught; 2 semesters).
1997 – 2011	Group Theory (7 semesters).
1997 – 2002	Computer Applications in Physics Research and Teaching (elective; 6 semesters).
1998 – 2003	Courses in the interdisciplinary programme *Sciences and Culture*: Introduction to Computers (60 h); Microcomputer Applications (74 h); Advanced Computer Applications (40 h); Design and Creation of HTML Pages (40 h).

University of Ioannina — Graduate courses

2014 – 2018	Electronic Learning Environments and Applications in Physics Teaching, M.Sc. *New Technologies and Research in Physics Education* (4 semesters).
2014 – 2016	Educational Technologies and Applications in Physics Teaching, same programme (2 semesters).
2005 – 2014	Modern Technologies in the Service of Education (9 semesters) and Development of Distance-Learning Methods (9 semesters), same programme.
2004 – 2007	Classical Mechanics, M.Sc. in Meteorology, Climatology and Atmospheric Environment Physics (2 semesters).
2002 – 2005	Computational Methods in Physics (elective), M.Sc. in Physics (2 semesters).

Development of Educational Material for Distance Learning

- 24-hour podcast series for PSF60 (Quantum Physics), Hellenic Open University (2008).
- Production of WEBCAST material for PSF60 (Quantum Physics), Hellenic Open University (2007–08).
- Critical review and evaluation of WEBCAST material for PSF50 (Classical Physics), Hellenic Open University (2007–08).
- Development of multiple-choice question banks for FYE34 (Classical Physics II), Hellenic Open University (2007–08).
- Complete digital open courses *Classical Electrodynamics I* and *Mathematics and Physics with Computers* for the institutional Open Academic Courses platform (`opencourses.gr`), University of Ioannina (2012–15).

Research Supervision

2004 – present	Supervision of **15 M.Sc. theses** within the graduate programmes of the Department of Physics, University of Ioannina, and of numerous undergraduate diploma theses. Supervision of **one doctoral dissertation**; member of three-member advisory committees for M.Sc. theses and doctoral dissertations, and of seven-member doctoral examination committees.
1999 – 2009	Supervision of five diploma theses in the interdisciplinary programme *Sciences and Culture*, University of Ioannina.

Academic Leadership and Administration

Department and University of Ioannina

2022 – present	**Chair**, Internal Evaluation Group (OMEA), Department of Physics.
2013 – 2017	**Head**, Department of Physics.
2013 – 2017	Member of the Deanery, School of Sciences.
2013 – 2014	Member of the Senate of the University of Ioannina.

2012 – 2020	**Chair**, Graduate Studies Coordinating Committee, M.Sc. *New Technologies and Research in Physics Education*.
2012 – 2017	**Chair**, Undergraduate Curriculum Committee, Department of Physics.
2009 – 2013	**Deputy Head**, Department of Physics.
2007 – 2013	**Director**, Division of Theoretical Physics.
2006 – 2011	Member of the Research Committee of the University of Ioannina, representing the Department of Physics.
2002 – 2013	Member, Committee for the Evaluation of Teaching, Department of Physics.
1998 – 2013	Member, Study Guide Committee, Department of Physics; responsible for the online edition.

Information technology

2005 – 2015	**Scientific Director**, Network Operations Centre, University of Ioannina.
2005 – 2013	**Chair**, Computer Committee, Department of Physics (member 2000–2005).
2001 – 2011	Member, Scientific Committee of the Network Operations Centre, University of Ioannina.
1997 – 2017	Responsible for the website of the Department of Physics; system administrator and webmaster of the Division of Theoretical Physics (1997–2013).

Hellenic Open University

2021 – 2024	Member, Curriculum Committee (E.P.S.) of the M.Sc. programme *Data Science and Machine Learning*.

Scientific Networks and Conference Organisation

2018 – present	**President**, Southeastern European Network in Mathematical and Theoretical Physics (SEENET-MTP), `seenet-mtp.info`. Re-elected July 2025 for a further two-year term.
2017 – 2018	Member, Scientific Advisory Committee, SEENET-MTP.

Conference and school organisation

2019	**Main organiser**, BS2019: SEENET-MTP Balkan School on High Energy and Particle Physics, Ioannina, 3–10 June 2019.
2017	**Main organiser**, HEP 2017: Recent Developments in High Energy Physics and Cosmology, Ioannina, 6–9 April 2017.
2016	**Main organiser**, STRINGPHENO 2016: 15th Conference in the String Phenomenology Series, Ioannina, 20–24 June 2016.
2015	Organising Committee, PLANCK 2015: 18th International Conference from the Planck Scale to the Electroweak Scale, Ioannina.
2015	**Main organiser**, *Beyond the Standard Models of Particle Physics and Cosmology*, workshop in honour of K. Tamvakis, Ioannina, 24 May 2015.
2012, 2006	Organising Committee, HEP2012 and HEP2006: Recent Developments in High Energy Physics and Cosmology, Ioannina.

2005	Organising Committee, CORFU2005: Corfu Summer Institute on Elementary Particle Physics, and *The Quest for Unification: Theory Confronts Experiment*, Corfu.
2001	Organising Committee, South European School on Elementary Particle Physics, Corfu (EC 5th Framework, HPCF-2001-00052-01), and the 4th RTN meeting *Across the Present Energy Frontier*.
2000	Organising Committee, Annual Workshop of the Hellenic Society for the Study of High Energy Physics, Ioannina.

Editorial and Refereeing Activity

2021 – present	Editorial Board, *Particles*.
2020 – present	Editorial Board, *International Journal of Modern Physics A*.
2020 – present	Editorial Board, *Modern Physics Letters A*.
2008 – present	Referee for *Nuclear Physics B*, *Journal of High Energy Physics*, *International Journal of Modern Physics A*, *Modern Physics Letters A* and *Universe*.

Professional Memberships

1999 – present	Marie Curie Fellows Association.
1990 – present	Hellenic Society for the Study of High Energy Physics.
1986 – present	Union of Greek Physicists.

Languages and Computing Skills

Languages:	Greek (native), English (fluent), French (very good).
Programming:	Python, Mathematica, FORTRAN 77/90/95, C++.
Web & scripting:	HTML, PHP, PERL, Java, JavaScript.
Systems:	UNIX/Linux, Windows, macOS.
Tools:	TeX/LaTeX, Word, Excel, PowerPoint, CorelDraw.

Active use of Python and Mathematica in both research and teaching. Application of optimisation algorithms — genetic algorithms and quantum annealing — to the classification of theoretical models (see *Selected publications on computational methods and machine learning* below).

Research Impact

INSPIRE-HEP:	5,173 citations • h-index 36 `https://inspirehep.net/authors/991609?ui-citation-summary=true`
Google Scholar:	5,886 citations • h-index 39 `https://scholar.google.com/citations?user=7TJ7ZnAAAAAJ`
ORCID:	`https://orcid.org/0000-0001-7586-1700`
Summary:	71 articles in international peer-reviewed journals; 1 invited handbook chapter; 14 papers in international conference proceedings; 5 papers in science-education proceedings; translation and scientific editing of a two-volume university physics textbook (3rd and 4th Greek editions).

Selected Invited Talks and Seminars

Jul 2025	*On the Quest for the Standard Model Among String Theory Vacua*, BWAM25: SEENET-MTP Workshop and Assessment Meeting, Bucharest, Romania.
Dec 2023	*On Non-Supersymmetric String Models*, Xmas Theoretical Physics Workshop, Athens.
Sep 2023	*On Three-Generation Super No-Scale Models in Heterotic String Theory*, CORFU2023, Corfu.
Aug 2023	*String Model Building Using Quantum Annealers*, BW2023 SEENET-MTP Meeting, Vrnjačka Banja, Serbia.
Aug 2023	*On String Model Building in the Free-Fermionic Formulation*, Department of Physics, University of Niš, Serbia.
Sep 2022	*On 3-Generation Non-Supersymmetric Heterotic String Models*, CORFU2022, Corfu.
Sep 2021	*Non-Supersymmetric String Phenomenology*, SEENET-MTP Workshop BW2021, Belgrade, Serbia.
Jul 2021	*On Non-Supersymmetric String Models and the Decompactification Problem*, Crete Center for Theoretical Physics, University of Crete.
Jun 2021	*On Non-Supersymmetric String Phenomenology*, Nuclear and Particle Physics Seminar, University of Athens.
Oct 2019	*On Non-Supersymmetric String Phenomenology*, Assessment Meeting of the SEENET-MTP–ICTP Project NT-03, ICTP, Trieste, Italy.
Sep 2019	*On Non-Supersymmetric String Model Building*, CORFU2019, Corfu.
Mar 2018	*Aspects of Non-Supersymmetric String Phenomenology*, HEP 2018, Athens.
Mar 2017	*Aspects of Non-Supersymmetric String Phenomenology*, Department of Physics, University of Cyprus, Nicosia.
Jul 2014	*Top Quark Mass Coupling and the Classification of SO(10) Heterotic Superstring Vacua*, String Phenomenology 2014, Trieste, Italy.
Apr 2013	*On the Classification of Heterotic Superstring Models*, Division of Theoretical Physics, University of Liverpool, UK.
Jul 2010	*Towards Classification of $Z_2 \times Z_2$ Heterotic Superstring Vacua*, String Phenomenology 2010, Paris.
Jan 2007	*Classification of $Z_2 \times Z_2$ Heterotic Orbifold Models and Spinor-Vector Duality*, University of Liverpool, UK.
Jan 2006	*The Standard Model and its Extensions from Strings*, Centre de Physique Théorique, École Polytechnique, Palaiseau, France.
Dec 2001	*A Pati-Salam Scenario from Branes*, Inaugural Meeting of the RTN Network *Across the Present Energy Frontier*, Oxford, UK.
Dec 1993	*Construction and Phenomenological Analysis of Superstring Unified Models in the Free-Fermionic Formulation*, SISSA/ISAS, Trieste, Italy.

Selected Publications on Computational Methods and Machine Learning

A subset of the works listed below, highlighted for relevance; these items are not additional to the counts given above.

1. **"String Model Building on Quantum Annealers,"** S. A. Abel, L. A. Nutricati and J. Rizos, *Fortsch. Phys.* **71** (2023) 2300167. — Application of quantum annealing to large-scale optimisation problems.
2. **"Genetic Algorithms and the Search for Viable String Vacua,"** S. Abel and J. Rizos, *JHEP* **1408** (2014) 010. — Application of evolutionary algorithms in high-dimensional parameter spaces.

3. **“Towards Machine Learning in the Classification of $Z_2 \times Z_2$ Orbifold Compactifications,”** A. E. Faraggi, G. Harries, B. Percival and J. Rizos, *J. Phys.: Conf. Ser.* **1586** (2020) 012032. — Among the earliest applications of machine-learning techniques to the classification of string models.
4. **“MERLIN: A Portable System for Multidimensional Minimization,”** G. A. Evagelakis, J. Rizos, I. E. Lagaris and I. N. Demetropoulos, *Computer Physics Communications* **46** (1987) 401–415. — Development of a multidimensional optimisation system; a foundational numerical-analysis tool directly relevant to model-training methods.

Publications in International Peer-Reviewed Journals

1. **“The Flipped SU(5) × U(1) Model from Four-Dimensional Strings,”** I. Antoniadis and J. Rizos, Universe **11** (2025) no.2, 44 doi:10.3390/universe11020044
2. **“From free-fermionic constructions to orbifolds and back,”** I. Florakis and J. Rizos, JHEP **01** (2024), 151 doi:10.1007/JHEP01(2024)151 `[arXiv:2310.10438 [hep-th]]`.
3. **“String Model Building on Quantum Annealers,”** S. A. Abel, L. A. Nutricati and J. Rizos, Fortsch. Phys. **71** (2023) no.12, 2300167 doi:10.1002/prop.202300167 `[arXiv:2306.16801 [hep-th]]`.
4. **“Three-generation super no-scale models in heterotic superstrings,”** I. Florakis, J. Rizos and K. Violaris-Gountonis, Phys. Lett. B **833** (2022), 137311 doi:10.1016/ j.physletb.2022.137311 `[arXiv:2206.09732 [hep-th]]`.
5. **“Particle physics and cosmology of the string derived no-scale flipped SU(5),”** I. Antoniadis, D. V. Nanopoulos and J. Rizos, Eur. Phys. J. C **82** (2022) no.4, 377 doi:10.1140/epjc/s10052-022-10353-6 `[arXiv:2112.01211 [hep-th]]`.
6. **“Super no-scale models with Pati-Salam gauge group,”** I. Florakis, J. Rizos and K. Violaris-Gountonis, Nucl. Phys. B **976** (2022), 115689 doi:10.1016/j.nuclphysb. 2022.115689 `[arXiv:2110.06752 [hep-th]]`.
7. **“Cosmology of the string derived flipped $SU(5)$,"** I. Antoniadis, D. V. Nanopoulos and J. Rizos, JCAP **03** (2021), 017 doi:10.1088/1475-7516/2021/03/017 `[arXiv:2011.09396 [hep-th]]`.
8. **“Doublet-Triplet Splitting in Fertile Left-Right Symmetric Heterotic String Vacua,”** A. E. Faraggi, G. Harries, B. Percival and J. Rizos, Nucl. Phys. B **953** (2020), 114969 doi:10.1016/j.nuclphysb.2020.114969 `[arXiv:1912.04768 [hep-th]]`.
9. **“Aspects of nonsupersymmetric string phenomenology,”** J. Rizos, Int. J. Mod. Phys. A **33** (2018) no.34, 1845002 doi:10.1142/S0217751X18450021.
10. **“Classification of left–right symmetric heterotic string vacua,”** A. E. Faraggi, G. Harries and J. Rizos, Nucl. Phys. B **936** (2018) 472 doi:10.1016/j.nuclphysb.2018.09.028 `[arXiv:1806.04434 [hep-th]]`.
11. **“Classification of Standard-like Heterotic-String Vacua,”** A. E. Faraggi, J. Rizos and H. Sonmez, Nucl. Phys. B **927** (2018) 1 doi:10.1016/j.nuclphysb.2017.12.006 `[arXiv:1709.08229 [hep-th]]`.
12. **“Wilsonian dark matter in string derived Z' model,”** L. Delle Rose, A. E. Faraggi, C. Marzo and J. Rizos, Phys. Rev. D **96** (2017) no.5, 055025 doi:10.1103/PhysRevD.96.055025 `[arXiv:1704.02579 [hep-ph]]`.
13. **“A Solution to the Decompactification Problem in Chiral Heterotic Strings,”** I. Florakis and J. Rizos, Nucl. Phys. B **921** (2017) 1 doi:10.1016/j.nuclphysb.2017.05.002 `[arXiv:1703.09272 [hep-th]]`.
14. **“Chiral Heterotic Strings with Positive Cosmological Constant,”** I. Florakis and J. Rizos, Nucl. Phys. B **913** (2016) 495 doi:10.1016/j.nuclphysb.2016.09.018 `[arXiv:1608.04582 [hep-th]]`.
15. **“The 750 GeV di-photon LHC excess and extra Z's in heterotic-string derived models”** A. E. Faraggi and J. Rizos, Eur. Phys. J. C **76** (2016) no.3, 170 `[arXiv:1601.03604 [hep-ph]]`.
16. **“A light Z′ heterotic-string derived model,”** A. E. Faraggi and J. Rizos, Nucl. Phys. B **895** (2015) 233 `[arXiv:1412.6432 [hep-th]]`.
17. **“Proton Stability in $SU(5) \times U(1)$ and $SU(6) \times SU(2)$ GUTs”** A. E. Faraggi, M. Paraskevas, J. Rizos and K. Tamvakis, Phys. Rev. D **90** (2014) 015036 `[arXiv: 1405.2274 [hep-ph]]`.
18. **“Genetic Algorithms and the Search for Viable String Vacua”** S. Abel and J. Rizos, JHEP **1408** (2014) 010 `[arXiv:1404.7359 [hep-th]]`.
19. **“Classification of Flipped $SU(5)$ Heterotic-String Vacua,”** A. E. Faraggi, J. Rizos and H. Sonmez, Nucl.

Phys. B **886** (2014) 202–242 [arXiv:1403.4107 [hep-th]].

20. **“Top quark mass coupling and classification of weakly-coupled heterotic superstring vacua”** J. Rizos, Eur. Phys. J. C **74**, 2905 (2014) [arXiv:1404.0819 [hep-ph]].
21. **“Shock waves as branes with throats”**J. Rizos and N. Tetradis, JHEP **1302** (2013) 112 [arXiv:1210.4730 [hep-th]].
22. **“String Derived Exophobic $SU(6) \times SU(2)$ GUTs”** L. Bernard, A. E. Faraggi, I. Glasser, J. Rizos and H. Sonmez, Nucl. Phys. B **868** (2013) 1 [arXiv:1208.2145 [hep-th]].
23. **“Classicalization as a tunnelling phenomenon**", J. Rizos, N. Tetradis and G. Tsolias, *JHEP* **1208** (2012) 054 [arXiv: 1206.3785 [hep-th]].
24. **“Dynamical Classicalization**", J. Rizos and N. Tetradis, *JHEP* **1204** (2012) 110, [arXiv:1112.5546 [hep-th]].
25. **“On the dynamics of classicalization**", N. Brouzakis, J. Rizos and N. Tetradis, *Phys. Lett.* **B708** (2012) 170-173, [arXiv:1109.6174 [hep-th]].
26. **“Top Quark Mass in Exophobic Pati–Salam Heterotic String Model**", Kyriakos Christodoulides, Alon E. Faraggi, John Rizos, *Phys. Lett.* **B702** (2011) 81-89 [arXiv: hep-ph/1104.2264]
27. **“Classification of Heterotic Pati–Salam Models**", B. Assel, K. Christodoulides, A. E. Faraggi, C. Kounnas and J. Rizos, *Nucl. Phys.* **B844** (2011) 365-396 [arXiv: hep-th/1007.2268].
28. **“Hierarchical Neutrino Masses and Mixing in Flipped-SU(5)**", J. Rizos and K. Tamvakis, *Phys. Lett.* **B685** (2010) 67-71 [arXiv: hep-th/0912.3997]
29. **“Exophobic Quasi-Realistic Heterotic String Vacua**", B. Assel, K. Christodoulides, A. E. Faraggi, C. Kounnas and J. Rizos, *Phys. Lett.* **B683** (2010) 306 [arXiv:hep-th/0910.3697].
30. **“Spinor-Vector Duality in Heterotic SUSY Vacua**", Tristan Catelin-Jullien, A. E. Faraggi, C. Kounnas and J. Rizos, *Nucl. Phys.* **B812** (2009) 103-127 [arXiv:hep-th/0807.4084].
31. **“Spinor-vector duality in N=2 heterotic string vacua**", A. E. Faraggi, C. Kounnas and J. Rizos, *Nucl. Phys.* **B799** (2008) 19-33 [arXiv: hep-th/07120747].
32. **“Spinor - vector duality in fermionic $Z_2 \times Z_2$ heterotic orbifold models”** A. E. Faraggi, C. Kounnas and J. Rizos, *Nucl. Phys.* **B774** (2007) 208-231 [arXiv: hep-th/0611251].
33. **“Renormalization effects on neutrino masses and mixing in a string-inspired $SU(4) \times SU(2)_L \times SU(2)_R \times U(1)_X$ model**", T. Dent , G.K. Leontaris, A. Psallidas, J. Rizos, *Phys. Rev.* **D75** (2007) 065027 [arXiv: hep-ph/0603228].
34. **“Chiral family classification of fermionic Z(2) x Z(2) heterotic orbifold models**" A. E. Faraggi, C. Kounnas and J. Rizos, *Phys. Lett.* **B648** (2007) 84-89 [arXiv: hep-th/0606144].
35. **“A D-brane inspired $U(3)_C \times U(3)_L \times U(3)_R$ model,”** G. K. Leontaris and J. Rizos, Phys. Lett. B **632**, 710 (2006) [hep-ph/0510230].
36. **“Gauge coupling and fermion mass relations in low string scale brane models”** D. V. Gioutsos, G. K. Leontaris and J. Rizos, *Eur. Phys. J.* **C45** (2006) 241-247 [arXiv:hep-ph/0508120].
37. **“Fermion masses and proton decay in string-inspired $SU(4) \times SU(2)^2 \times U(1)_X$”** T. Dent, G. Leontaris and J. Rizos, *Phys. Lett.* **B605** (2005) 399-405 [arXiv:hep-ph/0407151].
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